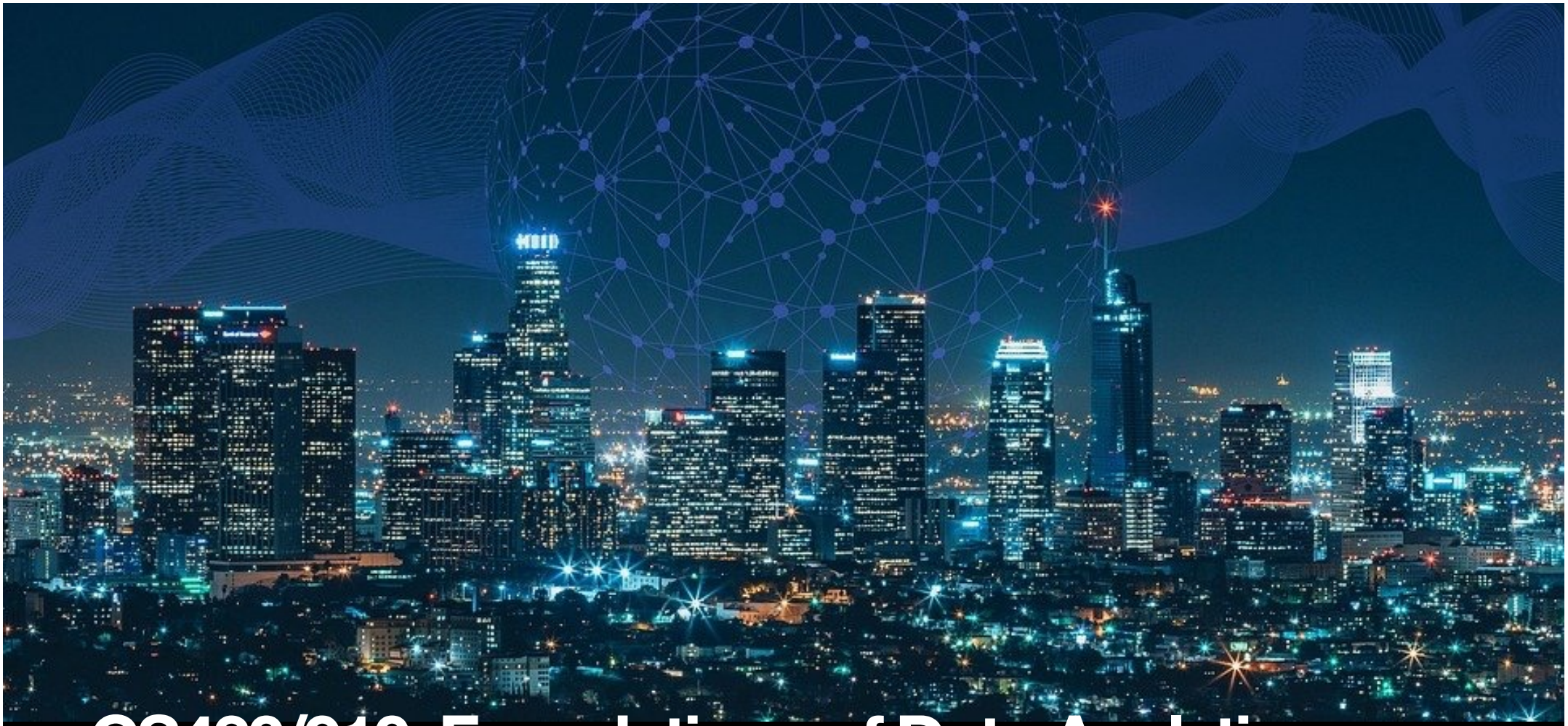

CS430/910: Foundations of Data Analytics

Ethical Practices in Data Science | Dr Ayse Sunar

We will cover

- Importance and impact of ethics in data science
- Case study of ethics in data science
- Techniques/tools/technologies to avoid ethical issues
- Ethical dilemmas in data science

Learning outcomes

Students will be able to;

- explain importance of ethics in data science
- discuss ethics in the context of data science
- comprehend and implement techniques/tools to avoid ethical issues
- create discussions on ethical dilemmas with cause/effect.

Understanding Ethics

Ethics is;

- What?

Ethics is not;

- What not?

Source

Understanding Ethics

Ethics is;

- about understanding and applying principles that help determine what is right or wrong - lying is unfair and harmful.
- being considerate - impact of your actions.
- about following professional codes - treating all patients with care and confidentiality.
- about debating moral dilemmas and trying to find out the most ethical action.

Ethics is not;

- about personal choices, emotions (personal feelings), social customs (related to a group of people, not all) - I like chocolate ice cream over strawberry.
- only about legal obligations – being legal does not mean ethical, e.g. racial segregation laws in the late 1800s.

Ethics in Data Science – Resource

- The guide in the following link has been developed jointly by the Royal Statistical Society (RSS) Data Science Section and the Institute and Faculty of Actuaries (IFoA).

- Most of the content in this session is based on this resource:

https://rss.org.uk/RSS/media/News-and-publications/Publications/Reports%20and%20guides/A-Guide-for-Ethical-Data-Science-Final-Oct-2019.pdf

Ethics in Data Science

- Responsible and fair use of data, algorithms, and technology in a way that respects privacy, fairness, transparency, and accountability. It incorporates decision-making that considers the impact on individuals, society, and the environment. The following is not a comprehensive list for ethical guidance:
  - Privacy –
  - Fairness –
  - Transparency –
  - Accountability –
  - Social impact –
  - Informed consent –
  - Security –

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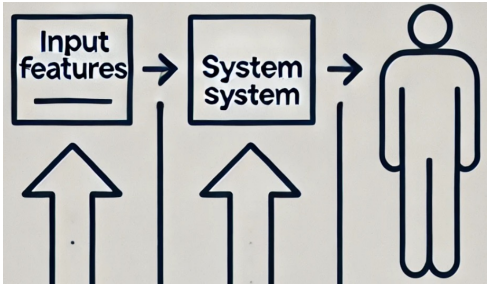

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  - Informed consent – informing users on how their data will be used
  - Security – take any measures for securing data to prevent any harm

Ethics - Social Impact

- Data science can be both beneficial or detrimental to the public.
- We need to understand the potential impact of our work to the society.
- We need to look for opportunities to deliver public good.

Ethics - Social Impact

How do you think data science can help society?

Ethics - Social Impact

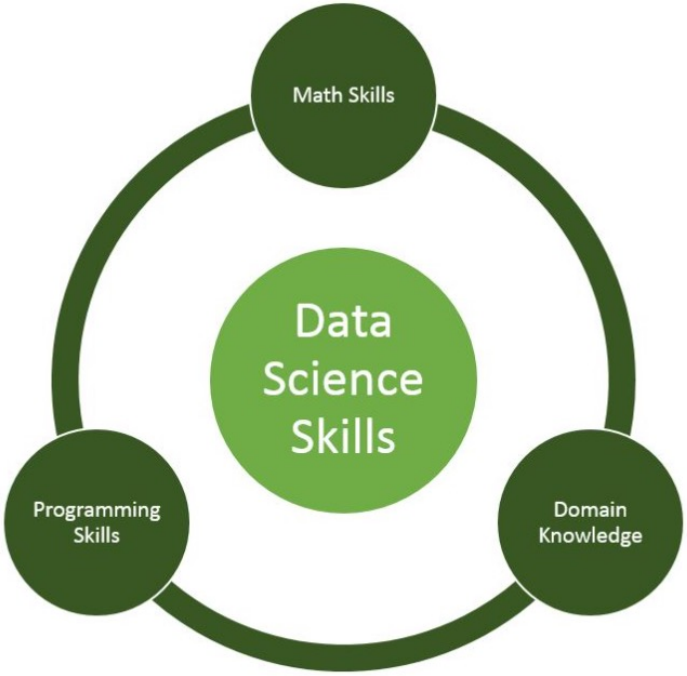
- Seek to improve public wellbeing
  - Data science can be used to analyse large datasets of medical records to identify patterns that predict the onset of diseases like diabetes or heart disease - preventive measures and reduction in healthcare costs.
  - Data science can help predict how climate change will affect different areas helping governments and organizations - prepare for natural disasters.
  - Data from traffic sensors can be analysed to optimize traffic flow in real-time - reduce congestion and pollution in urban areas.

Ethics – Professional Competence

- Seek to minimise risks by complying with the best industry and professional practices
- We need to pay attention to professional competence to comply with the ethical data science practices

Ethics – Professional Competence

How to comply with the best industry and professional practices?



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Ethics – Professional Competence

- Seek to minimise risks by complying with the best industry practices
  - A data scientist working in healthcare ensures that all data processing complies with the General Data Protection Regulation (GDPR) in Europe – anonymise patient data to protect privacy.
  - They follow best practices in data science by using version control and documenting their code - enable reproducibility so that others can verify the results.
  - In response to changes in the underlying data, the data scientist periodically retrains and updates the model – maintain the relevance and accuracy of the model.

Ethics – Trustworthiness

- Preserving and increasing public trust in data science is important for the adoption of data-driven technologies.
- Building trustworthy data science involves clear communication, transparency in methodologies, and accountability for the outcomes, as part of the ethical data science practices.

Ethics – Trustworthiness

Why do we need to preserve or increase trustworthiness in data science?

Ethics – Trustworthiness

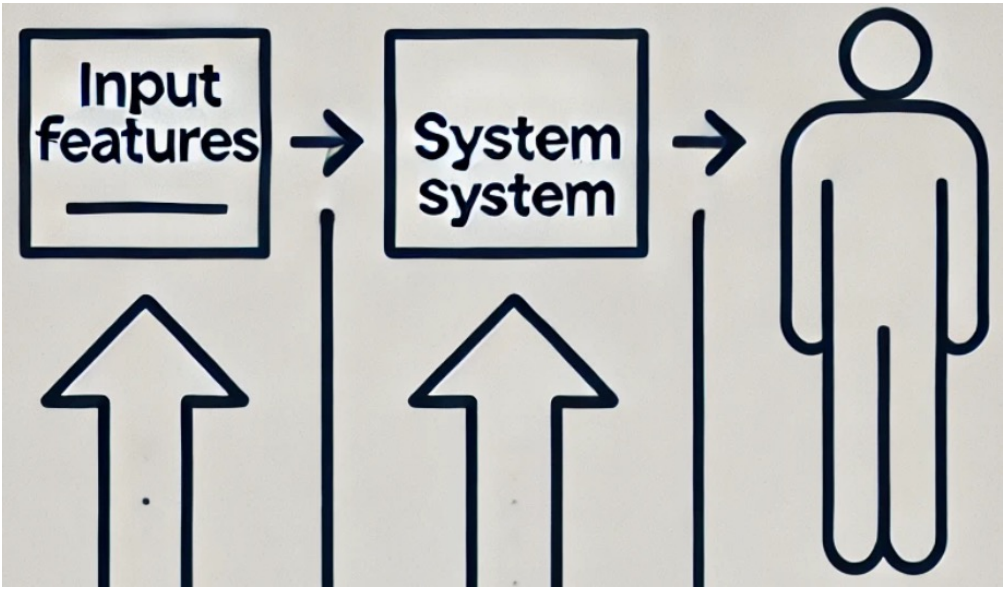
- Building public trust enables the adoption of data science.
  - Consulting ethical guidelines from professional bodies – adhering to widely recognised ethical standards.
  - Opting for interpretable models and documenting the choice of model – stakeholders understand how decisions are being made.
  - Proactively engaging with the public to address project concerns about privacy or bias – fostering trust and accountability.

Ethics – Accountability

- An ethical aspect is to take responsibility for a data-driven outcome and improve the process – maintaining accountability.
- We need to monitor and control when we let data-driven models make decisions – the level of oversight.
  - How much do we monitor and control what AI models do?

Ethics – Accountability

How to maintain accountability and oversight?



Ethics – Accountability

- Maintaining accountability and oversight brings trust and transparency.
  - We need to maintain oversight of automated AI solutions – to mitigate potential errors or biases, e.g. recruiters to review top recommendations of the automated hiring systems before a final decision.
  - We need to define operational constraints – e.g. limiting self-driving capabilities to certain type of roads or weather conditions. regularly reviewed to ensure safety and reliability.
  - Publishing accountability reports – detailing algorithms used, data sources, the outcome of their decisions, to be reviewed by external auditors to ensure transparency and adherence to ethical standards.

Ethics – Accountability

- An example: an entire HR team was terminated after a manager discovered that their automated system had mistakenly rejected all job applications, including his own.

Synopsis
A manager uncovered a major flaw in his company's applicant tracking system (ATS) that automatically rejected all job candidates, including his own CV under a fake name. The error stemmed from the system incorrectly filtering for outdated AngularJS skills instead of the required Angular expertise. After reporting the issue, the entire HR team was fired for relying too heavily on the flawed system without reviewing applications manually.

Ethics – Avoid Harm

- An ethical aspect is to think broadly how data can cause harm – to avoid harm.

Ethics – Avoid Harm

How to avoid harm, including financial loss, reputation damage, discrimination, etc.?

Ethics – Avoid Harm

- Avoiding harm requires transparency, integration of privacy safeguards, ethical risk management, education, etc. in data science.
  - We need to use ethically sourced data through transparent and informed consent – to avoid potential reputation damage or financial loss.
  - Teams need to create a framework for ethical risk management – e.g., continuous auditing and compliance with fairness standards to avoid discrimination.
  - Teams need to organise workshops to educate workforce to identify risks of harm and unfairness – e.g., including potential biases in machine learning models, fairness in data collection and processing.

Summing up - importance of ethics

- Protecting privacy and personal data – exposing private health information!
- Avoiding harm and discrimination – biased hiring and facial recognition systems!
- Ensuring transparency and accountability – trust issues in data-driven systems!
- Promoting fairness and equality – prioritising wealthier white patients over minority lower-income patients with greater medical needs!
- Complying with legal and regulatory standards – penalty of non-compliance to GDPR, up to EUR 20m!
- Building trust and credibility – selling data to third-parties and tens of unknown callers!
- Long term sustainability – unsustainable use of resources contributes to environmental degradation and climate change. Training an AI model can emit 626000 lbs of CO_2, nearly five times the lifetime emissions of an average American car.

Case Study – Cambridge Analytica (CA)

- Summary: Misuse of personal data for political purposes.
  - Leading to widespread public outcry and legal consequences.
  - Use of Facebook data without user consent to influence voter behaviour in political campaigns, including 2016 US presidential election and UK's Brexit referendum.
  - The scandal broke in 2018, when Cambridge Analytica harvested the personal data of millions of Facebook users without their consent to create psychographic profiles to manipulate voter behaviour.

Case Study – CA Methodology

- Methodology
  - Data collection
    - A third-party app, namely "this is your digital life" collected personal data of 87 million Facebook users.
    - 270k users directly consented, but app was able to collect from friends.
    - Data contained personal profiles, friend networks, likes, behavioural metrics, etc.
  - Psychometric profiling
    - CA used data science techniques to develop psychometric models categorising people's personalities (OCEAN model).
  - Micro-targeting
    - Using these psychographic profiles, CA created personalised political advertisement targeted at predicted vulnerabilities. For example, individuals deemed neurotic might be shown ads emphasising security threat to influence their decision making.

Case Study – CA Violations

- Ethical and legal violations
  - Lack of informed consent – users are not aware their data was used for political purposes.
  - Manipulation of voter behaviour – targeting people based on emotional or psychological vulnerabilities, potentially altering their votes.
  - Breach of trust – Facebook faced public backlash for failing to protect user data and for allowing third-party apps to access personal information without adequate oversight.
  - Violation of data protection laws - in 2019, Facebook was fined $5 billion by the U.S. Federal Trade Commission (FTC) for privacy violations, and Cambridge Analytica shut down following the backlash.

Case Study – CA Key Implications

- Key implications for data science
  - Transparency and Consent – ethical failures of the lack of transparency and informed consent.
  - Data Protection and Privacy – this scandal led to stricter controls on how personal data is collected, stored and used (GDPR).
  - Manipulation and Fairness – data scientists must consider the potential for harm when designing algorithms and models, particularly in sensitive areas such as politics, affecting democratic processes.
  - Accountability and Oversight - this case demonstrates the need for stronger oversight and accountability in data science projects.
  - Public Trust - The scandal significantly damaged public trust in data-driven technologies and platforms like Facebook. Organizations must maintain public trust by adhering to ethical principles.

Avoiding ethical issues – Techs/Tool

- Data anonymisation – ?
- Data minimisation –
- Differential privacy –
- Encryption –
- Diverse training data –

Avoiding ethical issues – Techs/Tool

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- Data minimisation – Collecting only the data that is necessary for the project, limiting privacy risks.
- Differential privacy – Adding noise to data so that data cannot be readily revealed.
- Encryption – protect sensitive information from unauthorized access.
- Diverse training data – training data used for machine learning must be representative of the population to avoid biased models, e.g., resampling techniques.

Data Anonymisation

- Binning technique to provide ranges, e.g. age 34 to age_group_31_50
- Mapping technique for a more generalised labelling, e.g., India/China to Asia
- Using libraries like faker

Fairness interpretation

- Calculate Precision, Recall, and F1 Score for each Gender: After making predictions, compute these metrics for each gender separately. In Python, you can use classification_report from sklearn which will give these scores for each class (for example, race or gender).

- Compare the Metrics: Check for large discrepancies between the metrics for race/gender. Significant differences indicate that the model may be biased.

Analysing precision, recall, and F1 score separately for different gender/race allows you to understand how the model performs for each group. Large discrepancies suggest the model is unfairly treating one compared to the other, and steps should be taken to mitigate this bias, such as resampling or algorithmic adjustments like fairness constraints.

Ethical dilemmas in Data Science

Can AI models ever be truly unbiased? Why or why not?

- Discuss with your peers for a few minutes. Search online and provide examples for the below aspects during discussions:
  - Data bias – unbalanced dataset, minority classes
  - Human factor – model related choices and decisions are made by human actor.
  - Algorithmic bias – one algorithm may prioritise a specific pattern over others.
  - Contextual bias – a model can perform differently in different environmental conditions, which may lead to biased outcomes.

Ethical dilemmas in Data Science

Can AI models ever be truly unbiased? Why or why not?

It is unlikely that AI can ever be fully unbiased. However, with responsible development and continuous monitoring, the goal should be to minimize biases as much as possible to ensure fairer outcomes.

Questions?

Any Questions?